



RIVER VALLEY HIGH SCHOOL

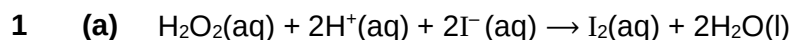
H2 CHEMISTRY

Suggested Solutions for 2012 A Level H2 Chemistry Paper 1

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	C	21	D
2	C	22	B
3	D	23	B
4	B	24	D
5	C	25	C
6	A	26	A
7	D	27	C
8	C	28	B
9	A	29	B
10	D	30	A
11	D	31	B
12	C	32	D
13	B	33	A
14	A	34	B
15	C	35	B
16	D	36	B
17	D	37	D
18	C	38	A
19	C	39	B
20	C	40	B

Suggested Solutions for 2012 A Level H2 Chemistry Paper 2

General comments from Cambridge examiners: "Candidates should read each question carefully before answering it in order to decide exactly what that particular question requires. When they have finished their answers they should check them to see if they have met the requirements of the question and whether they have made any simple errors such as writing unbalanced equations or omitting the units of a calculated quantity."



- (b) To have an **acidic buffer solution** so that **$[\text{H}^+]$ remains relatively constant throughout** the experiment.

(Wrong: H^+ is required for the reaction to occur.)

- (c) **Pre-calculations**

For a time reading of significantly below 20 s, $[\text{H}_2\text{O}_2]$ stock solution should be above 1.5 mol dm^{-3} . Assume desired $[\text{H}_2\text{O}_2]$ to be 3.0 mol dm^{-3} .

To dilute the H_2O_2 provided from 7.5 mol dm^{-3} to 3.0 mol dm^{-3} ,

If a 250 cm^3 volumetric flask is used,

volume of $7.5 \text{ mol dm}^{-3} \text{ H}_2\text{O}_2(\text{aq})$ required

$$= \frac{3.0}{7.5} \times 250 = 100.00 \text{ cm}^3 \text{ (requires the use of a burette twice)}$$

If a 100 cm^3 volumetric flask is used,

volume of $7.5 \text{ mol dm}^{-3} \text{ H}_2\text{O}_2(\text{aq})$ required

$$= \frac{3.0}{7.5} \times 100 = 40.00 \text{ cm}^3 \text{ (requires a single burette fill. preferred.)}$$

Preparation of H_2O_2 stock solution

- Using a **50 cm^3 burette**, accurately measure out **40.00 cm^3 of the $7.5 \text{ mol dm}^{-3} \text{ H}_2\text{O}_2(\text{aq})$** into a **$100 \text{ cm}^3$ volumetric flask**.

(Do NOT use a measuring cylinder / beaker / conical flask to measure out the 40.00 cm^3 . The concentration of the H_2O_2 stock solution needs to be accurately known.)

- Top up with distilled water **until the mark**. Use a dropper to add distilled water towards the end to ensure accuracy.
- Cover the flask and **shake well** to ensure the solution is well mixed.
- Label this flask as H_2O_2 stock solution.

Iodine clock experiments

- 1 Using a **25 cm³ measuring cylinder**, measure **25 cm³ of H₂O₂ stock solution** and pour it into a **250 cm³ conical flask**.
- 2 Using a **25 cm³ measuring cylinder**, measure **25 cm³ of 1.0 mol dm⁻³ CH₃COOH** and pour it into the **same conical flask**.
- 3 Using a **50 cm³ measuring cylinder**, measure **50 cm³ of solution A**.
- 4 Pour the 50 cm³ of solution **A rapidly** into the conical flask and **start the stopwatch at the same time**. (or when about half of solution **A** has been added)

(There is a need to show urgency in starting the timer.)

- 5 **Swirl** the conical flask 5 times to ensure complete mixing.
- 6 **Stop the stopwatch** when the solution turns **from colourless to dark blue**.
- 7 **Record the time taken** in table 1.
- 8 **Repeat** steps 1 to 6 for reaction mixtures 2 to 5, using **a 25 cm³ measuring cylinder** to measure the **volume of water** required and pour it into the conical flask before step 3.

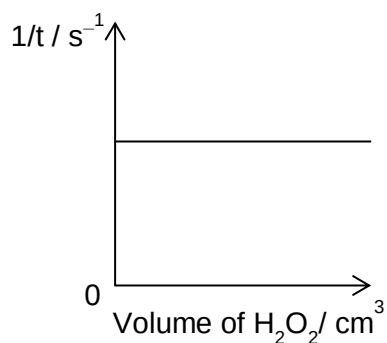
Table 1:

Rxn mix	Vol of soln <u>A</u> / cm ³	Vol of H ₂ O ₂ / cm ³	Vol of distilled water / cm ³	Vol of CH ₃ COOH / cm ³	Time taken for dark blue colour to appear / s	Reaction rate, $\frac{1}{t}$ / s ⁻¹
1	50.0	25.0	0.0	25.0		
2	50.0	20.0	5.0	25.0		
3	50.0	15.0	10.0	25.0		
4	50.0	10.0	15.0	25.0		
5	50.0	5.0	20.0	25.0		

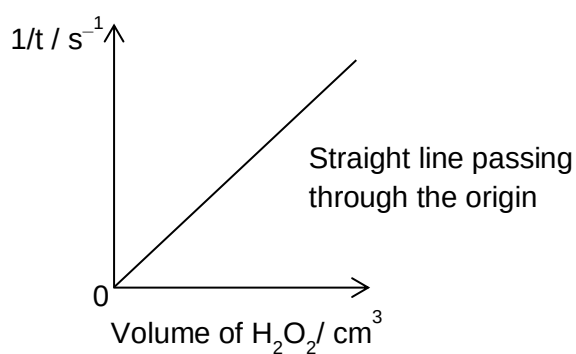
Determining the order of reaction with respect to H₂O₂

- 1) Plot a graph of reaction rate, $1/t$, (in s⁻¹) against volume of H₂O₂ used (in cm³).
- 2) Check the shape of the graph to determine the order of reaction with respect to H₂O₂:

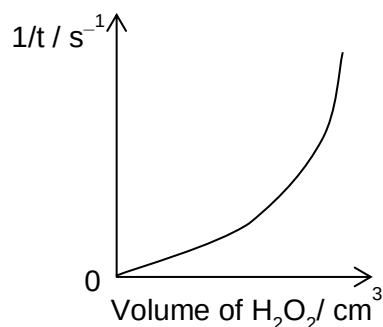
Zero order:



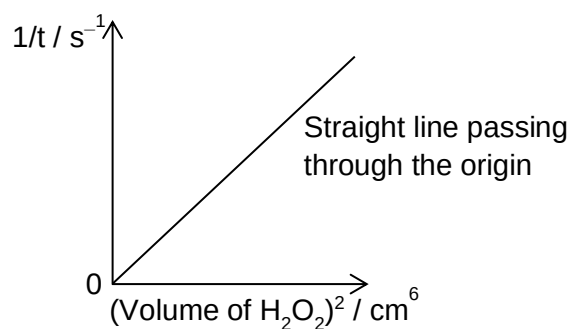
1st order:



2nd Order?



2nd Order:



- (d) H_2O_2 at such high concentration of 7.5 mol dm^{-3} is corrosive. Use of gloves is necessary to prevent direct contact of chemicals with the skin when carrying out the experiment.

(Reject: Chemicals used are flammable.)

- 2 (a) (i) Alkanes
- (ii) In the car engine, nitrogen from the air will react with oxygen to form NO.
- (iii) Proteins / Amino acids (cysteine) [3]
- (b) (i) $\Delta H^\ominus = \Delta H_f^\ominus(\text{SO}_3) - \Delta H_f^\ominus(\text{SO}_2)$
 $= -396 - (-297)$
 $= -99.0 \text{ kJ mol}^{-1}$
- (ii) $\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$
 $= -99 \times 10^3 - 298(-93.5)$
 $= -7.11 \times 10^4 \text{ J mol}^{-1}$ [4]
- (c) (i) $\Delta G^\ominus = -2.303RT \log K_c$
 $-7.11 \times 10^4 = -2.303(8.31)(298) \log K_c$
 $K_c = 2.93 \times 10^{12} \text{ mol}^{-1/2} \text{ dm}^{3/2}$
- (ii) The large magnitude of K_c shows that the equilibrium position of $\text{SO}_2 + \frac{1}{2}\text{O}_2 \rightleftharpoons \text{SO}_3$ lies largely to the right and the reaction is effectively complete.
- (iii) The formation of SO_3 from SO_2 has a very high activation energy.
- (iv) The catalyst is used to provide an alternative reaction pathway of lower activation energy so that SO_2 can be oxidised to SO_3 more quickly. [7]

[Total :14]

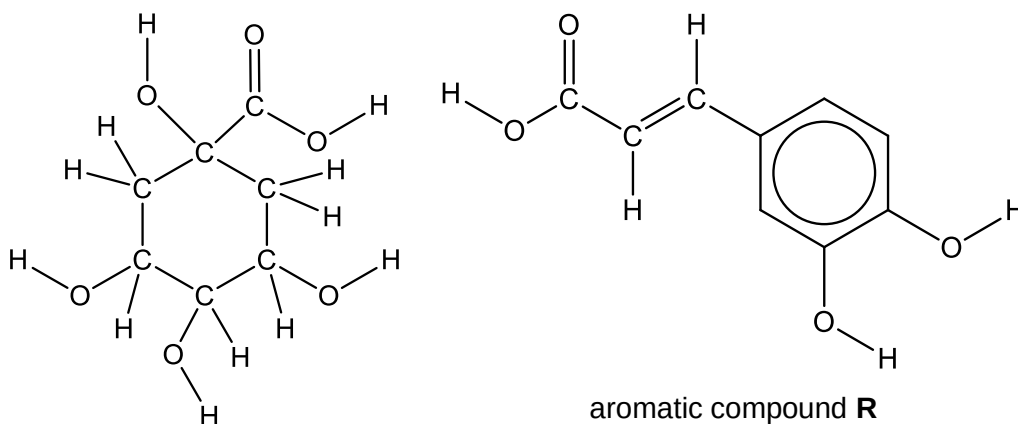
- 3 (a) (i) $K_{sp} = [Pb^{2+}] [Cl^-]^2 \text{ mol}^3 \text{ dm}^{-9}$
- (ii) Solubility = $4.7 / 278$
 $= 1.69 \times 10^{-2} \text{ mol dm}^{-3}$
 $K_{sp} = [Pb^{2+}] [Cl^-]^2 = (1.69 \times 10^{-2}) (2 \times 1.69 \times 10^{-2})^2$
 $= 1.9 \times 10^{-5} \text{ mol}^3 \text{ dm}^{-9} \text{ (2 sf)}$
- (b) (i) Immediately: White ppt of AgCl
After 1h in daylight: White ppt turns grey (AgCl decomposes to grey Ag)
- (ii) $AgCl (s) + 2NH_3 (aq) \rightarrow [Ag(NH_3)_2]^+ (aq) + Cl^- (aq)$
White ppt of AgCl dissolves in excess aq NH_3 to give a colourless complex of $[Ag(NH_3)_2]^+$.
- (iii) $PbCl_2 (s) + 2 Cl^- (aq) \rightarrow PbCl_4^{2-} (aq)$
White ppt of PbCl₂ dissolves in conc HCl to give a colourless complex of $PbCl_4^{2-}$.
- (c) $[Cu(H_2O)_6]^{2+} (aq) + 4Cl^- (aq) \rightleftharpoons CuCl_4^{2-} (aq) + 6H_2O (l)$
Blue Yellow
When concentrated HCl is added, equilibrium position shifts to the right to produce yellow complex $CuCl_4^{2-}$ according to Le Chatelier's Principle.
When water is added, equilibrium position shifts to the left to produce blue $[Cu(H_2O)_6]^{2+}$. Ligand exchange has occurred.
- (d) (i) $HS^-(g) \rightleftharpoons H^+(aq) + S^{2-} (aq) - (1)$
In an acidic solution, the presence of a high $[H^+]$ shifts the equilibrium position of (1) to the left, resulting in a low $[S^{2-}]$.
Hence, only CuS which has the lowest K_{sp} value will be precipitated.
- (ii) NiS has the highest K_{sp} value, hence it is the hardest to precipitate.
In an alkaline solution, the OH^- reacts with H^+ , reducing $[H^+]$. Equilibrium position of (1) will hence shift to the right, increasing $[S^{2-}]$ to a sufficiently high level for NiS to be precipitated.
- (iii) ZnS has a K_{sp} value that is between CuS and NiS. Therefore, $[S^{2-}]$ needed to form ZnS ppt will be intermediate between CuS and NiS. Hence, at least a neutral solution is required to produce a large enough $[S^{2-}]$. $[S^{2-}]$ in an acidic solution will be too low.

- 4 (a) (i) tertiary alcohol, secondary alcohol, carboxylic acid, ester, phenol

(ii) 4

[6]

(b)



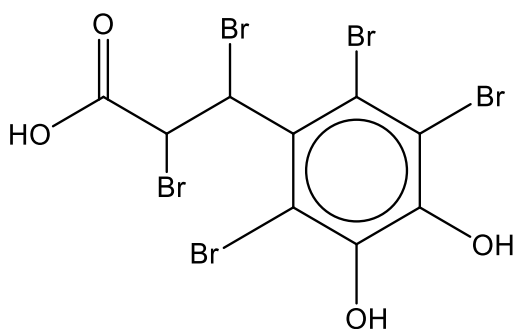
aromatic compound **R**

Non –aromatic compound

[2]

(Reject: structures of sodium salts)

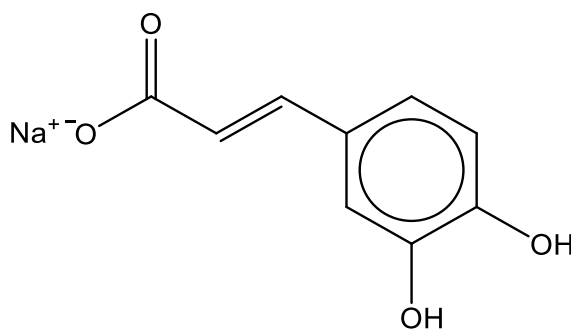
(c)



(Each of the two phenolic –OH groups would activate the ring, giving rise to substitution of two/three –Br in the ring.)

[2]

(d) (i)



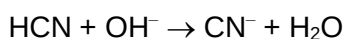
- (ii) The carboxylic acid group in **R** is a stronger acid than the phenol functional groups. Hence, only the carboxylic acid group is able to react with NaHCO₃, a weak base, to give one Na⁺ in **S**.

[3]

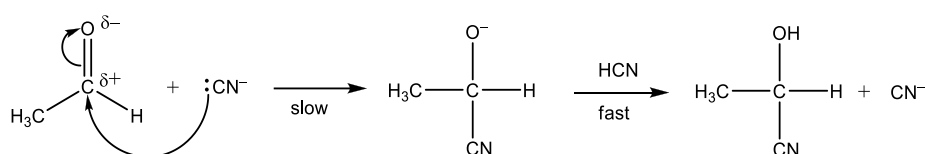
- 5 (a) (i) Reagent: HCN
 Conditions: Small amount of NaOH(aq) *OR*
 Small amount of NaCN(aq)
 Type of reaction: Nucleophilic addition

(ii) Generation of nucleophile:

- When HCN and a small amount of NaOH(aq) is used:



- When HCN and a small amount of NaCN(aq) is used:

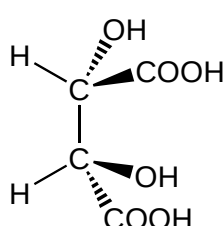
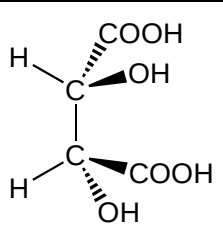
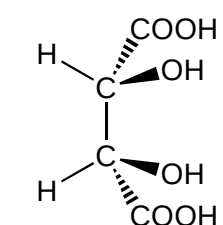


[5]

- (b) (i) Only **one enantiomer** of the lactic acid molecule is present in milk (i.e. all molecules are of the same chirality), hence it can rotate the plane of polarised light.
- (ii) The **carbonyl carbon** is sp^2 hybridised. Thus, there is a **trigonal planar** arrangement about the carbonyl carbon, allowing the **CN⁻ nucleophile** to **attack on either side of the plane with equal probability**. This results in a 50:50 **mixture of both enantiomers** being formed, and the **racemic** mixture does not rotate the plane of polarised light.

[4]

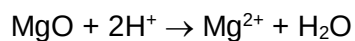
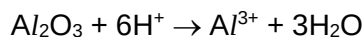
(c)

molecule	explanation	
A		Molecules possess two chiral carbons and no plane of symmetry. Hence, they are mirror images of each other that are non-superimposable (i.e. enantiomers of each other) Note: Structure of A and B can be interchanged.
B		
C		Although the molecule has two chiral carbons, there is a plane of symmetry present. Hence, the mirror images are superimposable.

[3]

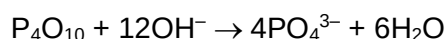
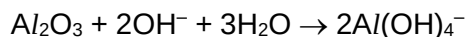
Paper 3

- 1 (a) Like MgO, Al_2O_3 can react with acids.



(Note: Both Al_2O_3 and MgO CANNOT dissolve nor react with H_2O .)

Like P_4O_{10} , Al_2O_3 can react with alkalis.

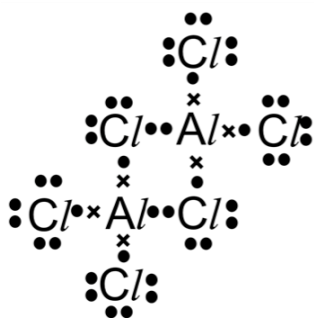


Al^{3+} has a **high charge density** and **polarises the O^{2-} ion**. Hence, Al_2O_3 is an **ionic** oxide with **covalent character** and is **amphoteric**.

(The question is concerned about Al_2O_3 , not $\text{Al}^{3+}(\text{aq})$. Hence, it is important to restrict your explanation to the interaction of the Al^{3+} ion with the O^{2-} ion, and NOT bring up the polarising and weakening of O-H bonds in H_2O molecules.)

[5]

- (b) (i)



- (ii) Trigonal planar

- (iii) According to Le Chatelier's Principle, when temperature is increased, the **equilibrium position** will **shift to the right** to favour the **endothermic reaction** to **absorb some of the heat added**. Yield of AlCl_3 will increase.

- (iv)

$$pV = \frac{m}{M}RT$$

$$\text{Average } M \text{ (Molar mass)} = \frac{mRT}{pV} = \frac{1.50(8.31)(500)}{(1.16 \times 10^5)(250 \times 10^{-6})}$$

$$= 215 \text{ g mol}^{-1}$$

$$\text{Average } M_r = 215 \text{ (No units)}$$

- (v) $215 = 267(1 - y) + 133.5y$

Solving,

$$x = \underline{0.610}, \quad y = \underline{0.390}$$

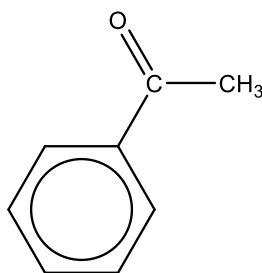
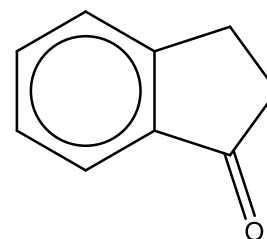
(vi) Partial pressure of Al_2Cl_6
 $= 0.610 \times (1.16 \times 10^5) = \underline{7.08 \times 10^4 \text{ Pa}}$

Partial pressure of $AlCl_3$
 $= 0.390 \times (1.16 \times 10^5) = \underline{4.52 \times 10^4 \text{ Pa}}$

(vii) $K_p = \frac{p(AlCl_3)^2}{p(Al_2Cl_6)} = (4.52 \times 10^4)^2 / 7.08 \times 10^4 = \underline{2.89 \times 10^4 \text{ Pa}}$

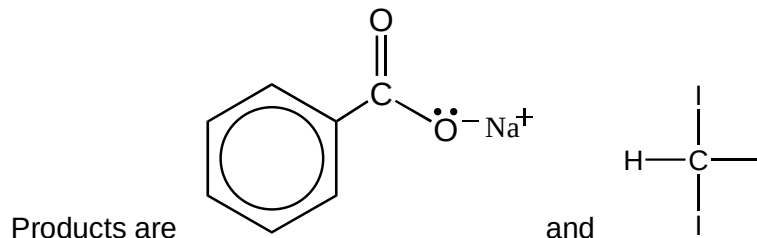
[11]

(c) (i)

**A****B**

(ii) Distinguishing test: Aqueous alkaline iodine, warm

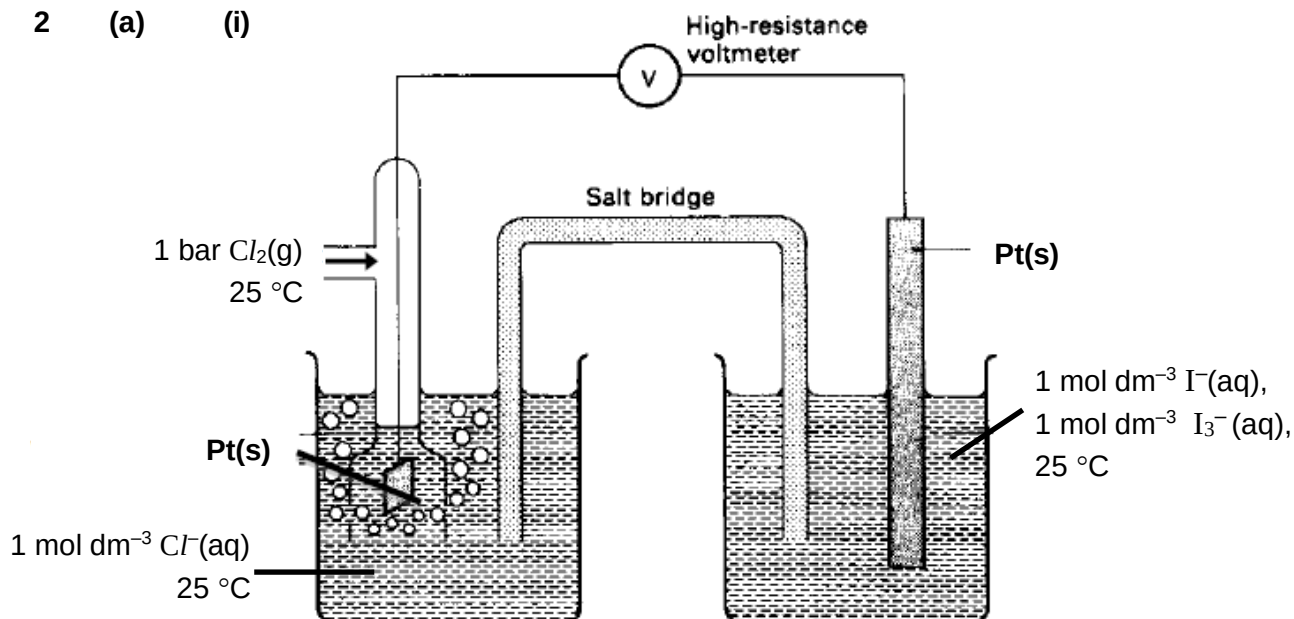
For **A**: Pale yellow ppt observed.



For **B**: Will not undergo iodoform reaction and no pale yellow ppt will be observed.

[4]

2 (a) (i)



(Note:

- It is given in the question that a solution containing I_3^- reacts as if it is a solution of I_2 in water.
- It is given in the question that only 1 electrochemical cell should be drawn.

Common mistakes:

- Use $I_2(s)$ electrode – it is non-conducting
- Pass $I_2(g)$ over Pt electrode – iodine is a solid at room temperature, although it sublimes readily.
- Missing salt bridge
- Include a battery in the circuit)

(Refer to Electrochemistry tutorial for more help)

(ii) **Measure the $E^\ominus_{\text{cell}}$** using the voltmeter. The **more positive** half-cell contains the **stronger oxidising agent**.

(More positive half-cell = More positive $E^\ominus$ = Reduced more easily = Stronger oxidising agent)

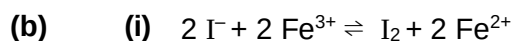
(Extra Physics information: To see which is the more positive half-cell during an experiment, you can look at the sign of the voltmeter reading.

If reading is +ve, it means that the positive terminal (red wire) is connected to the more positive half-cell, and the negative terminal (black wire) to the more negative half-cell.

If reading is –ve, it means that the positive terminal (red wire) is connected to the more negative half-cell, and the negative terminal (black wire) to the more positive half-cell.)

(iii) $Cl_2 + 2 I^- \rightarrow 2 Cl^- + I_2$

[7]



(ii) CN^- forms complexes with both Fe^{3+} and Fe^{2+} .

From the data booklet,

$$E^\ominus([\text{Fe}(\text{CN})_6]^{3-}/[\text{Fe}(\text{CN})_6]^{4-}) = +0.36\text{V} \quad \textbf{(least positive)}$$

$$E^\ominus(\text{Fe}^{3+}(\text{aq})/\text{Fe}^{2+}(\text{aq})) = +0.77\text{V}$$

$$E^\ominus(\text{I}_2(\text{aq})/\text{I}^-(\text{aq})) = +0.54\text{V}$$

The reduction of $[\text{Fe}(\text{CN})_6]^{3-}$ is less favourable than $\text{Fe}^{3+}/[\text{Fe}(\text{CN})_6]^{3-}$ cannot oxidise I^- under standard conditions.

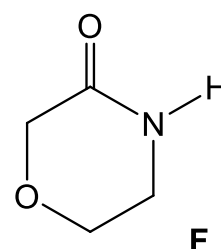
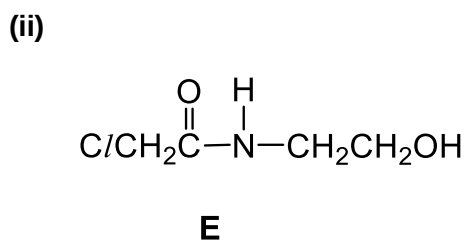
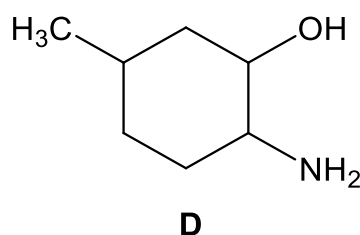
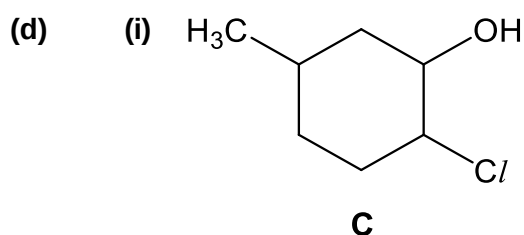
Therefore, the equilibrium position of (b)(i) shifts to the left.

(Note: CN^- forms complexes with both $\text{Fe}^{3+}(\text{aq})$ and $\text{Fe}^{2+}(\text{aq})$. Hence, adding CN^- causes both $[\text{Fe}^{3+}(\text{aq})]$ and $[\text{Fe}^{2+}(\text{aq})]$ to decrease. Therefore, CANNOT use the decrease in $[\text{Fe}^{3+}(\text{aq})]$ to explain the shift in equilibrium position.)

[3]

(c) C–Cl bond is stronger than C–I. Hence, it requires more energy to break the C–Cl bond for hydrolysis. Thus, $\text{C}_2\text{H}_5\text{Cl}$ reacts more slowly than $\text{C}_2\text{H}_5\text{I}$.

[1]



(amine group is more basic than alcohol group. This implies that lone of pair e^- on $-\text{NH}_2$ is more available for donation. Hence, $-\text{NH}_2$ is a better nucleophile and will react with the acid chloride to form an amide.)

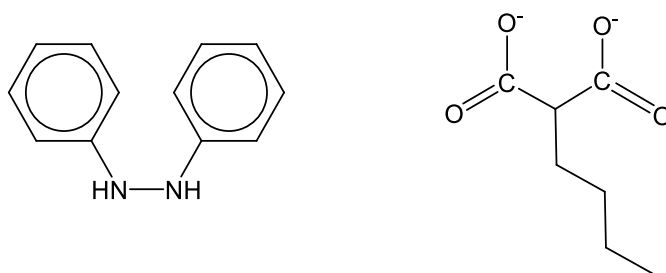
(Na will deprotonate $-\text{OH}$ group to $-\text{O}^-$ group. The $-\text{O}^-$ group will then undergo intramolecular $\text{S}_{\text{N}}2$ reaction and substitute the $-\text{Cl}$ group on the other end of the molecule to give a cyclic product, as mentioned in the question.)

[4]

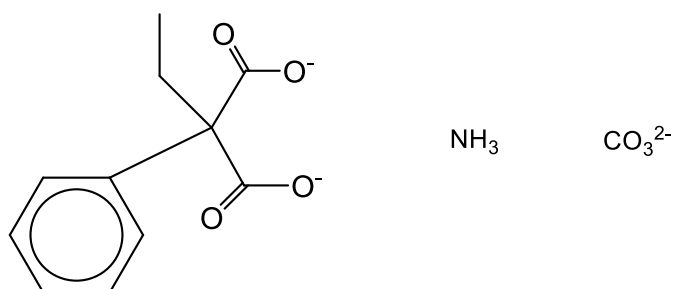
(e) (i) $\text{NaOH}(\text{aq})$, heat under reflux OR $\text{HCl}(\text{aq})$, heat under reflux

(ii) Products from alkaline hydrolysis of phenylbutazone:

[3]



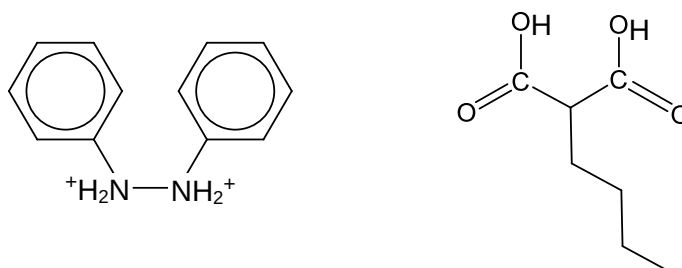
Products from **alkaline** hydrolysis of phenobarbital:



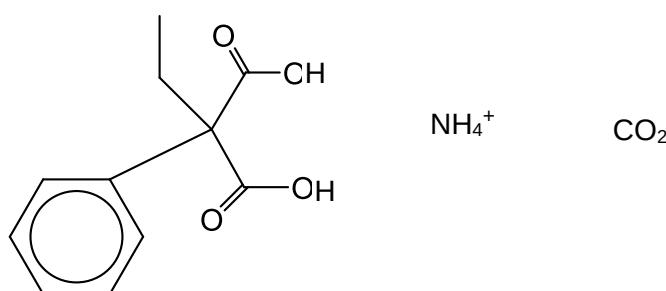
(Note: For hydrolysis of phenobarbital, CO(NH₂)₂ will not be obtained since it can further hydrolyse into NH₃ and CO₃²⁻.)

OR

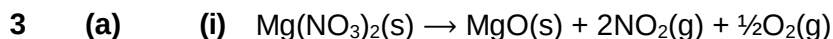
Products from **acidic** hydrolysis of phenylbutazone:



Products from **acidic** hydrolysis of phenobarbital:



(Note: For acidic hydrolysis of phenobarbital, H₂CO₃ will not be produced. Remember that H₂CO₃ will decompose readily to CO₂ and H₂O.)

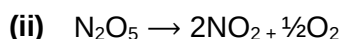
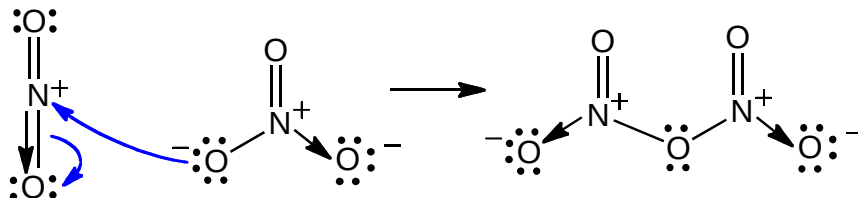
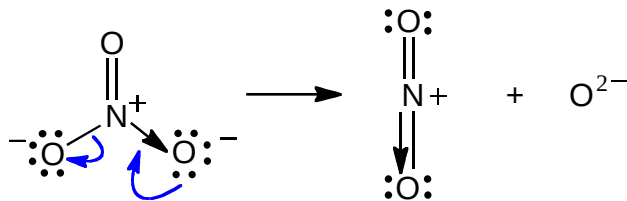


(ii) **Thermal stability** of Group II nitrates **increases** down the group.

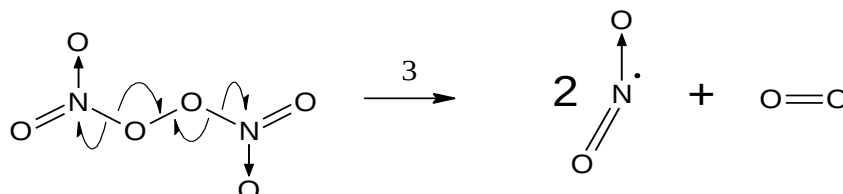
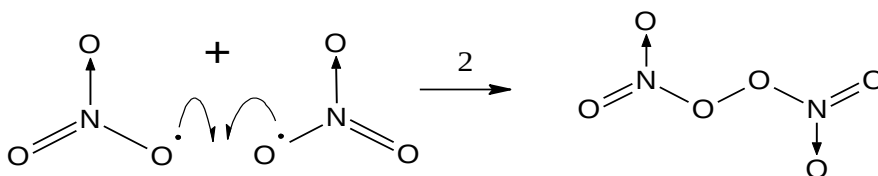
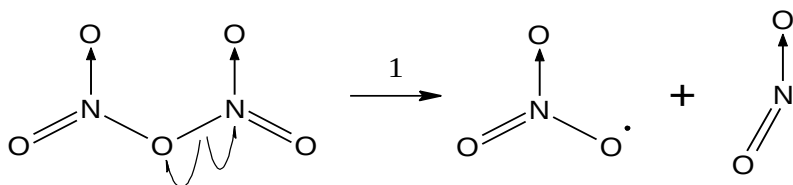
Down the group, **ionic radius of M^{2+} increases**, resulting in a **smaller charge density** and a **lower polarising power** of M^{2+} . Hence, the **N-O bond in the NO_3^- ion is weakened to a smaller extent** and **more energy** is required to break down the nitrate.

[3]

(b) (i)



(iii)



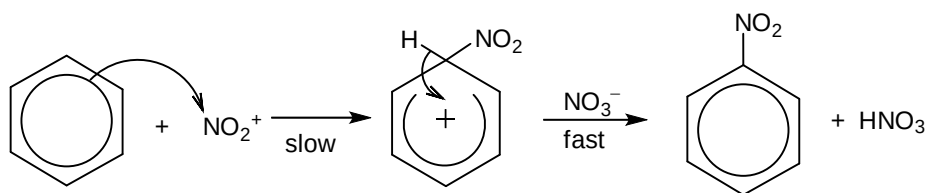
(Note: Use half arrows to show transfer of one single electron)

(iv) Since the reaction is overall **first order with respect to $[\text{N}_2\text{O}_5]$** , the **first step** is the **slowest** step since it is the only step that **involves only 1 molecule of N_2O_5** .

[7]

(c) (i) $\text{HNO}_3(\text{aq})$

(ii) Given that in solution, $\text{N}_2\text{O}_5 \rightarrow \text{NO}_2^+ + \text{NO}_3^-$



Key points

- ✓ Arrow from benzene ring to NO_2^+
- ✓ Arrow from C-H bond into the ring
- ✓ HNO_3 produced from NO_3^- and H^+

[3]

(d) (i) $\Delta H_r = \sum \Delta H_f(\text{products}) - \sum \Delta H_f(\text{reactants})$

$$+92.5 = \Delta H_f(\text{NO}_2(\text{g})) + \Delta H_f(\text{NO}_3(\text{g})) - \Delta H_f(\text{N}_2\text{O}_5(\text{g}))$$

$$\Delta H_f(\text{NO}_3(\text{g})) = +92.5 - (+33.2) + (+5.0) = \underline{+64.3 \text{ kJ mol}^{-1}}$$

(ii) $\Delta H_r = 2(\Delta H_f(\text{NO}_2(\text{g}))) - 2(\Delta H_f(\text{NO}_3(\text{g})))$
 $= 2(+33.2) - 2(+64.3) = \underline{-62.2 \text{ kJ mol}^{-1}}$

(Note: $\Delta H_f(\text{O}_2(\text{g}))$ is zero.)

[3]

(e) $\Delta G = \Delta H - T\Delta S$

ΔS is positive for both reactions since there is an **increase in number of moles of gaseous molecules**. Hence **$-T\Delta S$ is negative** for both reactions.

Since **ΔH is positive for reaction 1**, **ΔG is negative only at high temperatures** when **$-T\Delta S$ is more negative**.

Since **ΔH is negative for reaction 2**, **ΔG is negative at all temperatures**.

Hence, **reaction 2** has a **more negative ΔG** , and hence will be **more spontaneous**.

[4]

- 4 (a) Angle of deflection is proportional to charge/mass ratio.

Specie	Charge/mass ratio	Angle of deflection
H ⁺	+1/1	+15°
D ⁺	+1/2	+7.5°
T ⁺	+1/3	+5°
He ²⁺	+2/4	+5°

[3]

- (b) (i) If angle of deflection of R is +5°, then its charge/mass ratio is +1/3.

$$+1/3 = \text{charge}/12$$

$$\text{charge} = \underline{+4}$$

- (ii) Number of neutrons = 12 – 6 = 6

$$\text{Number of electrons} = 6 – 4 = \underline{2}$$

[3]

- (c) (i) Ionic radii in increasing order: Si⁴⁺ < Na⁺ < Cl[–] < P^{3–}

Na⁺ (0.095 nm) and Si⁴⁺ (0.041 nm) belong to an isoelectronic series. Si⁴⁺ has more protons, resulting in a higher nuclear charge. Since Na⁺ and Si⁴⁺ have the same total number of electrons, shielding effect is approximately the same. Thus, Si⁴⁺ has a higher effective nuclear charge, and the attraction on valence electrons by the nucleus of Si⁴⁺ is stronger, resulting in a smaller ionic radius.

(Wrong: Si⁴⁺ lost more electrons. Hence, remaining electrons are held more strongly by the Si⁴⁺ nucleus.)

Cl[–] (0.181 nm) and P^{3–} (0.212 nm) belong to another isoelectronic series. They have larger ionic radii than Na⁺ and Si⁴⁺ due to an additional shell of electrons.

Cl[–] has more protons, resulting in a higher nuclear charge. Since Cl[–] and P^{3–} have the same total number of electrons, shielding effect is approximately the same. Thus, Cl[–] has a higher effective nuclear charge, and the attraction on valence electrons by the nucleus of Cl[–] is stronger, resulting in a smaller ionic radius.

(Wrong: Greater negative charge results in more shielding effect)

- (ii) Melting points in increasing order: Cl₂ < P₄ < Na < Si.

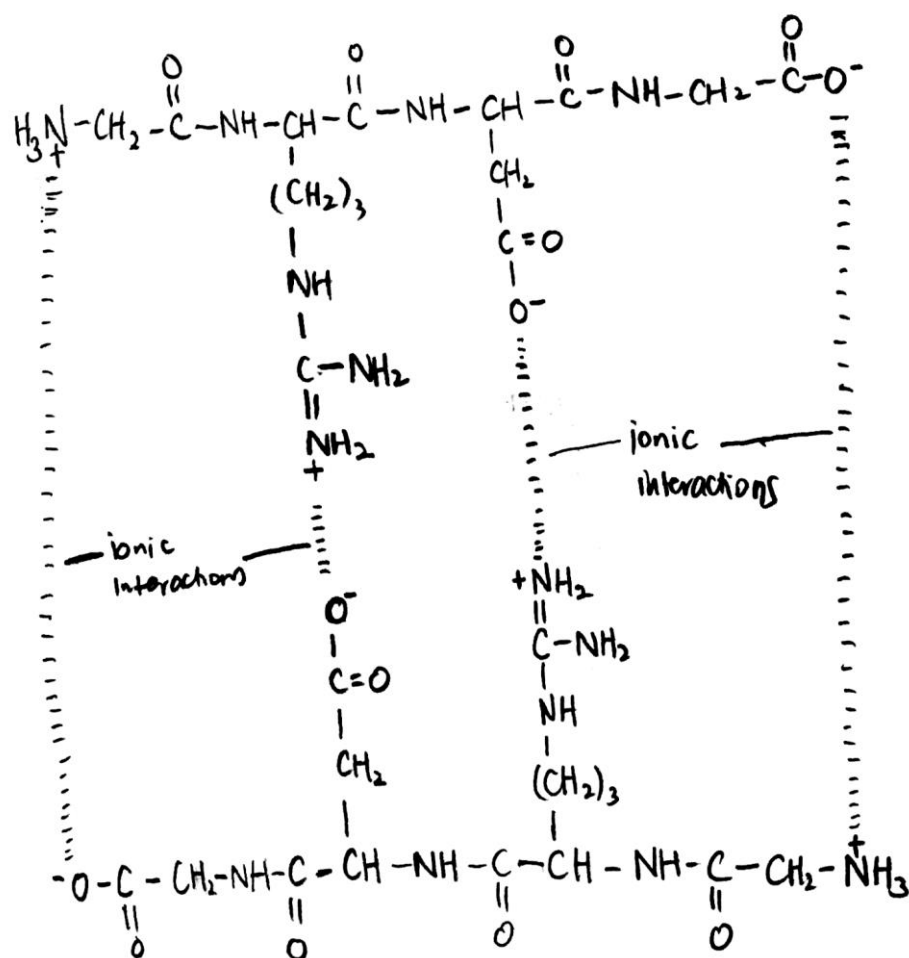
Chlorine and phosphorus exist as Cl₂ and P₄ molecules respectively. Both have simple covalent structures and instantaneous dipole-induced dipole (id-id) interactions between their molecules. As P₄ molecule has more electrons than Cl₂ molecule, more energy is required to overcome the stronger id-id interactions between P₄ molecules as compared to that between Cl₂ molecules.

Sodium has a giant metallic lattice structure. Even more energy is required to overcome the stronger electrostatic attraction between Na⁺ ions and its sea of mobile valence electrons.

Si has a giant covalent structure. Highest amount of energy is required to overcome the strongest covalent bonds between Si atoms.

[6]

(d) (i) (Not in 9729 syllabus)



(You can choose to protonate any of the 3 nitrogen atoms on the arginine residue.)

(ii) (Not in 9729 syllabus)

The tertiary structures of proteins can be stabilised by hydrogen bonds, disulfide linkages or van der Waals' forces between the R groups of amino acid residues.

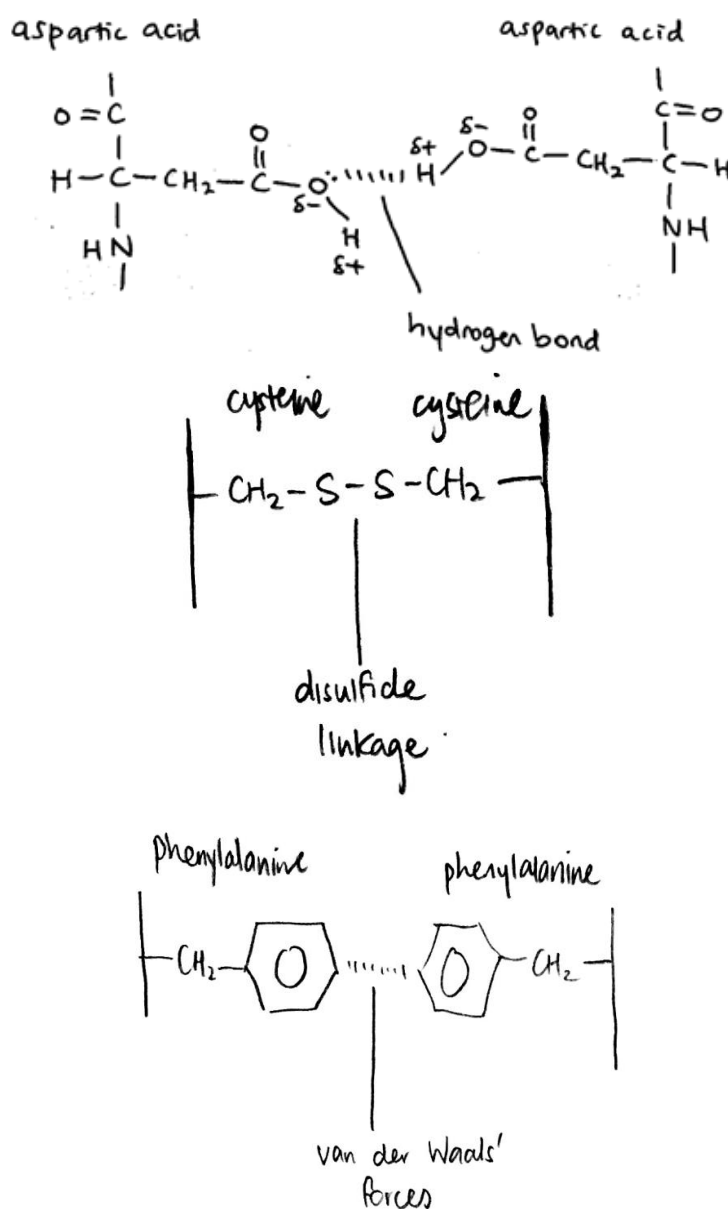
Hydrogen bonds can be formed between two amino acid residues whose R groups contain $-OH$ and/or $-NH_2$ groups.

Disulfide linkages can be formed between the $-SH$ groups of two cysteine residues.

Van der Waals' forces (NOT "Hydrophobic interactions") can be formed between two amino acid residues whose R groups are non-polar.

OR

Draw out:

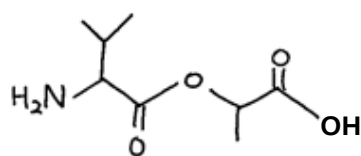
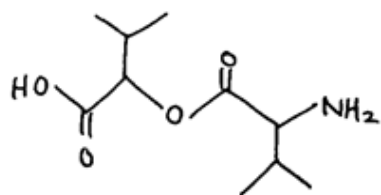


[5]

(Any two)

(e) (i) Ester

(ii)



[3]

[Total :20]

- 5 (a) Step 1: HCN, small amount of NaCN

OR

HCN, small amount of NaOH

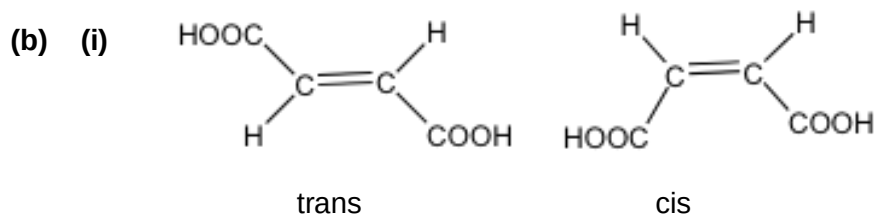
Step 2: Alcoholic KCN / NaCN, heat under reflux

Step 3: $\text{H}_2\text{SO}_4(\text{aq})$, heat under reflux



(Order of steps 1 and 2 can be interchanged.)

[5]



cis-trans isomerism

- (ii) Excess concentrated H_2SO_4 , heat (at 170°C)

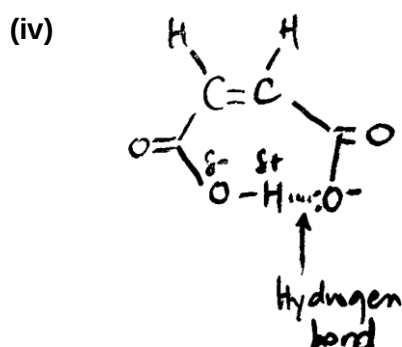
(Wrong: Excess concentrated $\text{H}_2\text{SO}_4(\text{aq})$)

OR

Al_2O_3 , heat (at 350°C)

- (iii) K produces the more stable mono-anion as it has a smaller first pK_a value and a larger second pK_a value.

(A larger second pK_a implies that the mono-anion formed after the first dissociation will not readily further dissociate again to give a di-anion and another H^+).



Key points

- Partial charges on O–H bond
- Lone pair on O^- group
- Dotted line
- Label of 'hydrogen bond'

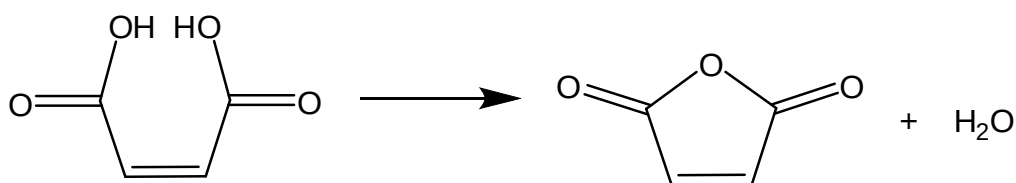
The *cis* mono-anion is able to form intramolecular hydrogen bond with the adjacent non-ionised COOH , thus making it more stable.

(O^- is attracted to δ^+ H. Hence, the mono-anion is less likely to react with a H^+ ion to give back the undissociated acid.)

[6]

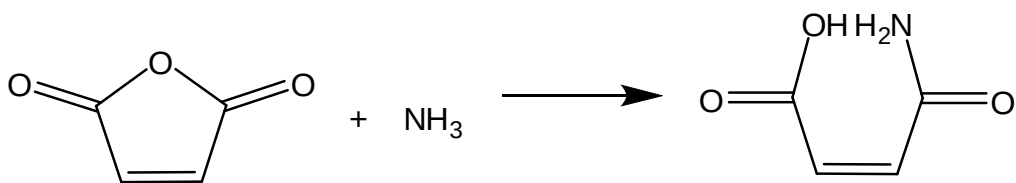
(c)

Information	Deduction
K undergoes <u>condensation</u> (loss of H ₂ O) upon heating to produce L	
L does <u>not</u> undergo <u>acid-metal reaction</u> with sodium metal	L does <u>not</u> contain an <u>-OH / -COOH group</u>
L does <u>not</u> undergo <u>condensation</u> with 2,4-DNPH	L does <u>not</u> contain a <u>carbonyl group</u> L contains an <u>ester</u> functional group
L reacts with ammonia to form M	M contains an <u>amide</u> functional group
M undergoes <u>acid-base reaction</u> with NaOH to give a salt	M contains <u>-COOH group</u>
M does <u>not</u> undergo <u>acid-base reaction</u> with HCl	M does <u>not</u> contain an <u>amine</u>
M undergoes <u>electrophilic addition</u> with Br ₂ (aq)	M contains <u>C=C</u> functional group



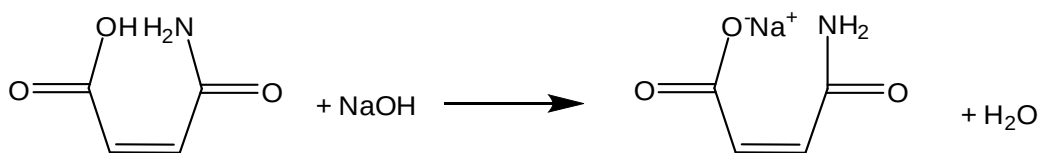
K

L

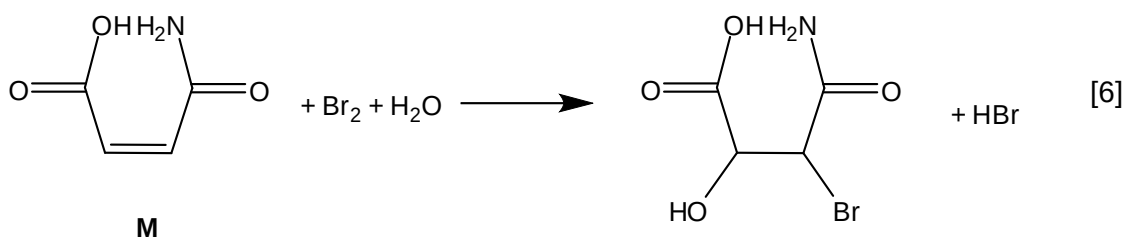


L

M

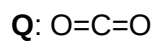
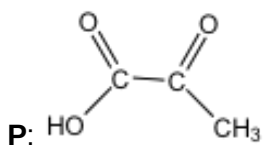
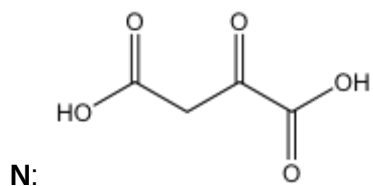


M



(Positions of –Br and –OH groups can be interchanged.)

(d)



[3]